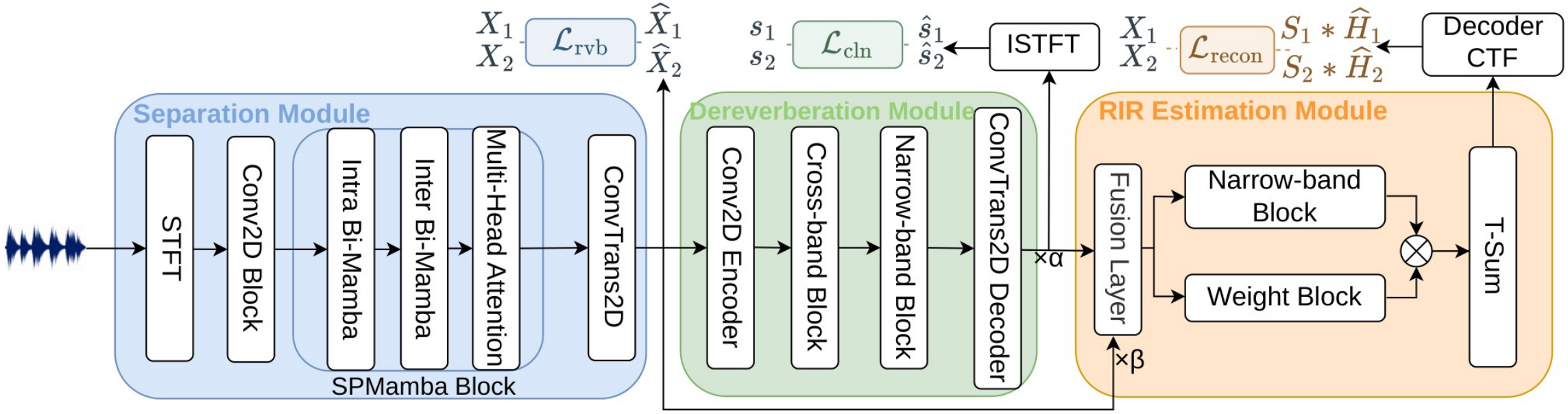
DAMSEP

Distance-Aware Monaural Source Separation using Multi-RIR Estimation

Wen Wen et al., arXiv:2609.29749v2 (2026)

Jointly estimates clean sources and source-specific RIRs from a monaural mixture.
DRR of each estimated RIR determines the relative near/far order.



Source: Wen et al., arXiv:2609.29749v2, Fig. 1 (cropped from the paper).

DAMSEP: Results

HETMIXR: 2-source mixtures, 8 kHz, 2,801 test mixtures

Source separation (dB, higher is better)

Model	Params	SI-SDRi ↑	SDRi ↑	SIR ↑	SAR ↑
TDANet	2.3M	8.08	7.94	17.95	9.33
SPMamba	6.1M	13.06	11.68	22.28	12.48
TF-Locoformer	15M	14.25	12.67	22.50	13.51
DAMSEP	**7.2M**	**14.90**	**13.24**	**24.29**	**13.91**

RIR estimation (lower is better; selected baseline)

Model	RIR-50	LSD (dB)	DRR MAE (dB)	C50 MAE (dB)
Rec-RIR	0.051	3.58	5.85	11.07
DAMSEP	**0.032**	**1.84**	**2.58**	**6.38**

Near/far accuracy: 99.11% | SI-SDRi: +0.65 dB vs. TF-Locoformer

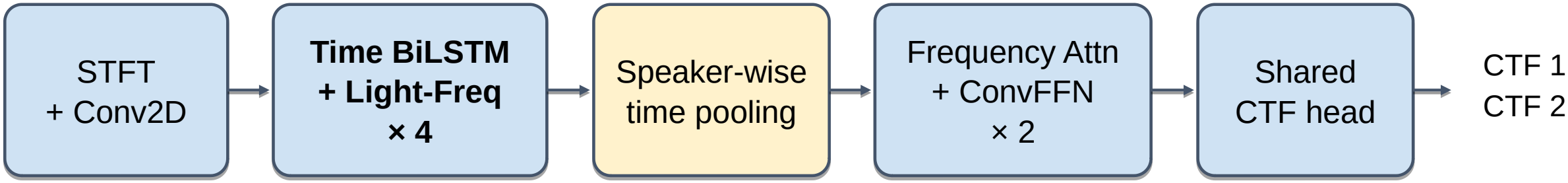
RIR-50: first-50-ms RMSE. Rec-RIR uses oracle single-source inputs; DAMSEP uses mixtures.

Source: Wen et al., arXiv:2609.29749v2, Tables 1–2.

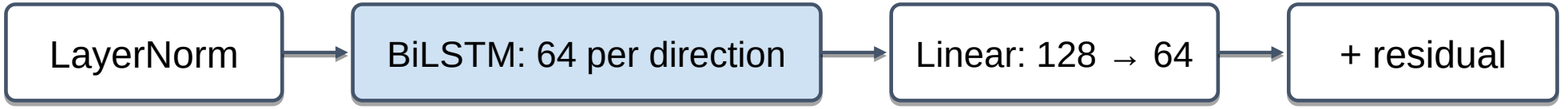
BLSTM-based CTF Estimator

Time-BiLSTM variant used in the previous CTF estimator comparison

Input: monaural mixture (16 kHz) Output: two source-specific complex CTFs



Time BiLSTM (independently at each frequency bin)



D = 64 | Pooling hidden = 32 | Attention = 4 heads | FFN = 128

STFT: FFT 512 / hop 256 | Output per source: 257 frequency bins × 60 CTF taps

Source: local TimeBiLSTM implementation and comparison configuration (16 kHz).